Whitney Learning Compass
From Curriculum Goals to Hands-On Discovery

Catapult · Learning Plan

Saved Draft — Not Revalidated
This plan was saved on Oct 23, 2026, 5:12 PM. Published program details and eligibility have not been rechecked. Confirm current information with the museum before booking. This local document cannot authorize a new AI workflow.

HISTORICAL PROGRAM SNAPSHOT · READ-ONLY

Catapult

At generation: Program Requirements Met (historical; not revalidated).

Students experiment with a rubber-band-powered catapult, varying band count and the position of the supplied ping-pong ball to explore force and motion.

SAVED PUBLISHED GRADES
3 4 5 6 7+

SAVED SUBJECTS
Design · Energy Transfer · Engineering · Force and Motion · History

SAVED PUBLISHED FEE
$14.00 per student

SAVED CLASS PROGRAM FEE
USD 336.00

SAVED WORKSHOP DURATION
90 minutes

BOOKING
Not confirmed

USD assumed for this Connecticut prototype; not explicitly published by the museum.

Historical program-fee subtotal only; transportation, meals and unverified additional charges are excluded. This is not a current quote.

Your class: Grade(s) 4 · 24 students

Learning goal: I want my fourth-grade students to explore force, motion, and engineering design through hands-on experiments.

Limits: USD 400 program-fee budget · 90 minutes maximum

Interests: Force and motion, engineering design, experimentation

Teacher-Edited Teaching Suggestions

Editable teaching suggestions, not official museum curriculum. Editing does not trigger model calls. Check the local saving status before leaving; printing provides a separate copy.

BEFORE

Prepare & Wonder

Build curiosity before the workshop.

Learning objectives

Recovered objective: compare real observations with a prediction.

Students will describe one step in an engineering-design process by comparing a prediction with observed results and explaining a possible reason for any difference.

Classroom preparation

Invite students to recall examples of pushes and pulls and how they can change an object’s motion.

Prepare a simple paper-and-pencil recording sheet with columns for trial or setup, prediction, actual observation, and possible explanation; leave observation fields blank until observations occur.

Review that any catapult use and handling during the program follow museum staff directions.

Background concepts

A force is a push or pull that can change an object’s motion.

The catapult activity varies band count and the position of the supplied ping-pong ball; students can compare what they observe across those changes.

Engineering design involves making a prediction, observing a result, and using evidence to consider what to change or explain.

Inquiry questions

How might changing the number of bands affect the catapult’s motion?

How might changing the ping-pong ball’s position affect what happens?

What evidence would help you compare two setups fairly?

DURING

Explore & Observe

Follow staff directions. Collect evidence, not invented results.

Observation prompts

If museum staff permit, attend to how the catapult's motion appears to change when the band count or ball position changes; record only what is actually observed.

Use a blank table: setup/change | prediction | actual observation | possible explanation. Leave the actual-observation column empty until the event is observed.

Follow museum staff directions for all catapult handling and procedures; do not adjust or operate equipment independently.

Hypothesis & prediction questions

Before a staff-directed comparison, what do you predict may happen if the band count changes? Label this as a prediction, not an observation.

What do you predict may happen if the ball's position changes? What makes you think so?

Critical-thinking prompts

Which factor changed between the setups you observed, and what appeared to stay the same?

Does the observed result support your prediction, differ from it, or leave the question open? What specific observation informs your thinking?

What other factor might affect the motion, and what would you need to observe before drawing a conclusion?

Evidence collection

Recovered activity: leave measurements blank until the actual visit.

Keep predictions separate from observed evidence, and note when an observation is unavailable or unclear rather than guessing.

Ask museum staff before recording or comparing any details not included in the staff-directed activity.

AFTER

Reflect & Extend

Connect actual observations to your learning goals.

Reflection & discussion

Which actual observation most helped you think about force and motion? Explain how it connects to the setup.

Where did an observation support or challenge your prediction? What evidence led you to that view?

What would you still want to observe before making a stronger claim about how a setup affects motion?

Follow-up classroom activity

Recovered offline edit: discuss only actual visit observations.

Learning assessment

Objective 1: Review each student's explanation for a connection between a changed setup feature (band count or ball position) and motion, supported by an actual observation. If none is available, assess the student's ability to state what evidence would be needed instead.

Objective 2: Check whether the student separates prediction from observed result and uses the observation to explain a possible outcome or difference; do not score hypothetical outcomes as empirical evidence.

Optional extension

Students can add a paper-and-pencil design reflection: identify one feature they would want to compare next, state a clearly labeled prediction, and list the actual observation that would be needed to evaluate it.

Teacher notes & actual observations

Leave observations blank until the visit occurs. Label predictions as predictions; do not treat them as collected evidence.

Teacher notes retained. No experiment has been conducted.

Questions to Confirm with the Museum

Saved questions; no answers or availability have been confirmed.

- What accessibility accommodations can you arrange?
- Are there any additional charges beyond the published per-student program fee?
- What dates are available for this program?
- What booking conditions and reservation steps apply?
- Can you accommodate our class of 24 students?
- Are there any special delivery or location arrangements to confirm?

Safety & Important Notes

Educator review required. Follow museum staff procedures; do not independently operate equipment or substitute projectiles. Use only approved activities and materials.

The museum's safety statement is conditional on using its supplied ping-pong ball; this record does not establish a general safety guarantee.

Use actual observations. Keep predictions separate from collected evidence.

Sources & Verification

Official museum program page ↗
https://www.eliwhitney.org/catapult

Original source review: 2026-10-21. This saved snapshot has not been revalidated.

Source information recorded at generation: program description (factual summary), published grades, subjects, workshop fee, duration.

Program-specific standards: Not published in the saved record.

▼ **Saved limitations and assumptions**

- Currency code: Confirm the currency code before a currency-sensitive budget check.
- Program-specific curriculum standards: Check for program-specific published standards or seek museum clarification; do not infer codes.
- Minimum class size: Confirm any minimum group size with the museum if required.
- Class capacity: Confirm workshop capacity with the museum if required.
- Accessibility arrangements: Confirm the class's relevant access requirements with the museum.
- Booking availability: Confirm dates and availability with the museum before planning a booking.
- Museum page update date: Not published. Our source review date is shown above.
- USD assumed for this Connecticut prototype; not explicitly published by the museum.

Teacher review pending. Teaching suggestions are not official museum curriculum. No experiment or student outcome is claimed to have occurred.

Independent educational prototype based on publicly available Eli Whitney Museum & Workshop program information. Not affiliated with, endorsed by, or operated by the museum.